



Database schema 12m

What is a database schema?

A database schema is developed during database design. It describes the overall logical structure of the database, organisation of data, constraints on the data and how the data is associated within the database structure.

For relational databases based on interrelated tabular structures, the schema describes the:

- names of the tables
- attributes included in each table
- relationships between attributes and tables.

In general, a relational database often includes multiple schemas that correspond to different users of the database with different privileges. A schema can be adjusted to suit the user's needs, and access levels can be set so that only specific tables are available, with restrictions on who can change the data.

The three-schema architecture

Data in a database is stored and presented in a highly organised form. It is organised based on user views, needs and expectations. These requirements are processed and developed by analysts and designers into logical structures that reflect the components of the users' enterprise and their relationship to the database. The logical constructions obtained in this way are then populated with more detailed descriptors that are relevant to the users' business, forming concrete data structures that are ready to accept the actual data.

[Skip to main content](#)

This way of describing database architecture is called the three-schema architecture (or the three-level architecture).

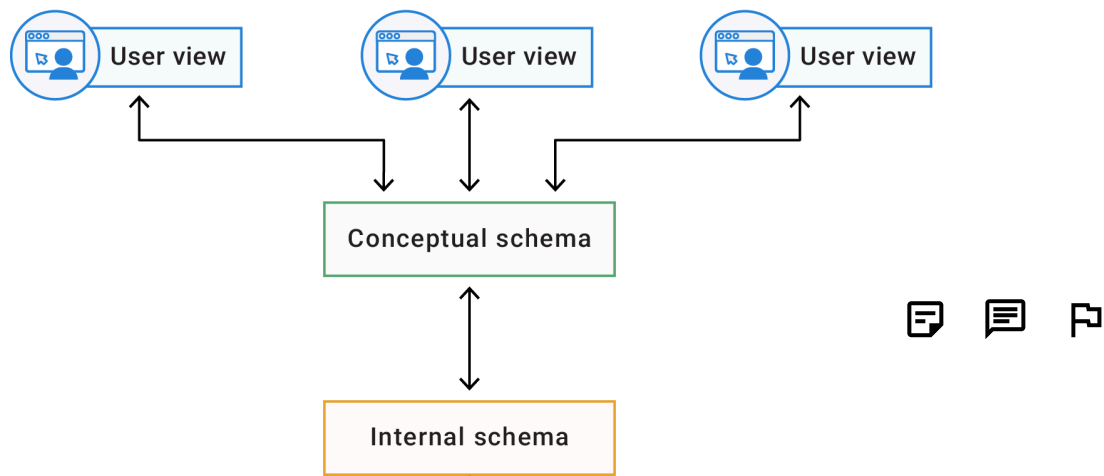
The three-schema architecture consists of an external schema, a conceptual schema and an internal schema. Let's look at each of these.

- **An external schema** represents the multiple user views of the data environment. In other words, an external schema includes several individual user views of the database data, defined according to the user's role within the organisation. Each user view is related to a particular part of the database.
- **A conceptual schema** represents the analyst's integrated interpretation of the whole organisational data (a global view of the data structure). In other words, a conceptual schema brings together different user views, creating a single and inclusive data definition.
- **An internal schema** develops from further analysis and design work on the database. This schema represents the adaptation of the conceptual model to a specific DBMS. An internal schema has two components:
 - **Logical schema**, which reflects the type of the chosen database technology
 - **Physical schema**, which defines the details of data storage and the ways of accessing data.



[Skip to main content](#)

Three schema architecture



The major feature of the three-schema architecture is that it promotes data independence.

Data independence is the characteristic that a modification of the schema at a lower level of a database system does not require a change in the higher-level schemas (only some mapping of data access between the schemas is required). Data independence helps you to keep data separated from the programs that use it. For instance, if a conceptual schema is modified (for example, by changing data format or logical data structure), the external schema does not need to be changed. Database applications also don't need any changes, as they interact only with the external schema. This type of data independence is called **logical data independence**.

In cases when an internal schema is modified (for example, by changing storage structures or storage devices), the conceptual schema does not need any changes. This type of data independence is called **physical data independence**.

The table below explains the difference between logical and physical data independence in more detail.

Type	Physical data independence	Logical data independence
------	----------------------------	---------------------------

Skip to main content

Data independence	► This is related to how the data is stored in the system. ► It is easy to retrieve the data. ► When it is compared to logical independence, it is easy to achieve physical data independence.	► This is related to how the data is stored in the system. ► It is easy to retrieve the data. ► When it is compared to logical independence, it is easy to achieve physical data independence. ► This is all about the structure or the changing data definition. ► Since the data is entirely dependent on the logical structure, it is difficult to retrieve the data directly with an application. ► When compared to physical independence, it is not easy to achieve logical data independence.
Flexibility	► Any change at the physical level does not require change at the application level. ► The modifications made at the internal level may or may not be needed to improve the performance of the structure.	► Any change at the physical level does not require change at the application level. ► The modifications made at the internal level may or may not be needed to improve the performance of the structure. ► The change at the logical level requires a change at the application level. ► The modifications made at the logical level are significant whenever the logical structure of the database is to be changed.

[Skip to main content](#)

Schema	It is related to the internal schema.	- It is related to the internal schema. - It is related to the conceptual schema.
--------	---	---



Dig Deeper

- Read *Three Schema Architecture* (TpointTech, n.d.) for a clear understanding of how the three-schema architecture in database management systems separates user views from physical data storage, enabling data independence, security, and flexibility in accessing and managing data.

Let's now look at the database schema for MySQL.



Thoughts on this page?

